Problem Chosen	2024	Team Control Number
D	MCM/ICM Summary Sheet	2416805

A Network Model of the Great Lakes with Accuracy and Robustness

Summary

The Great Lakes of the United States and Canada are the largest group of freshwater lakes in the world. Controlling water levels of the Great Lakes is difficult because of changing environmental conditions and conflicting interests. Our article establishes a network model of the Great Lakes and provides an approach to effectively manage lakes' water levels.

Before all the models are established, we make the **data preparation**. Then the **equivalent replacement model** of the lake is built, and finally we perform **fitting analysis** to obtain the seasonal rules of precipitation, runoff, evaporation and river flow.

For the first task, we are required to build a **network model** of the Great Lakes. Based on the real geographical location, a network graph was established with **lakes as nodes and rivers as edges**. We collected relevant literature and find that changes of lake's water volume are related to precipitation, runoff, evaporation, river inflow, and river outflow. We use these factors to establish **water conservation equations** that every nodes of the network must follow. The relationships between the water levels of different lakes is described by **river flow formulas**.

For the second task, in order to determine the **optimal water levels** of the Great Lakes, a **multi-objective programming model** is built. We fully consider the interests of various stakeholders and the stability of each lake. As a result, we calculate that the optimal water levels of the five Great Lakes in any month of the year. And the annual average optimal water levels of the five Great Lakes is obtained.

The third task asks for the control algorithm for two dams' outflow to maintain the optimal water levels. We propose the **PID feedback control algorithm** which can help to calculate the regulated flows of the two dams. Thereafter, we compare the water levels obtained using the PID feedback algorithm with the recorded water levels in 2017. It is found that the water levels controlled by the algorithm is closer to the optimal water levels. Finally, **sensitivity analysis** is performed. We improve the stability and efficiency of the regulation process by adjusting the combination coefficients of the PID equation. Except that, the changes in environmental conditions are taken into account. We conclude that our model is **robust** in the face of **abnormal environmental changes**.

For the fourth task, we need to focus on the stakeholders of Lake Ontario and develop plans to regulate Lake Ontario water levels accordingly. We propose **the satisfaction evaluation model** to measure the degree of satisfaction of the stakeholders. Then based on changes in the flow of the Ottawa River, we recommend a control plan for Lake Ontario. Finally, we use radar charts to show the changes in stakeholder satisfaction values before and after the implementation of the regulation plan, and conclude the superiority of the plan.

Keywords: Network model; Multi-Objective Programming Model; PID Feedback Control Algorithm; Sensitivity Analysis

Contents

1 Introduction

1.1 Problem Background

The Great Lakes, located at the junction of the United States and Canada, are the largest group of freshwater lakes in the world and are known as the "North American Mediterranean"(Shown in Figure 1). These five lakes and their interconnecting waterways form a vast watershed that covers many of the largest cities in both countries. The waters of these lakes are used for a variety of purposes, and as a result there are many stakeholders associated with the Great Lakes. How to balance the interests of these relevant stakeholders is an issue that still needs to be considered and resolved.

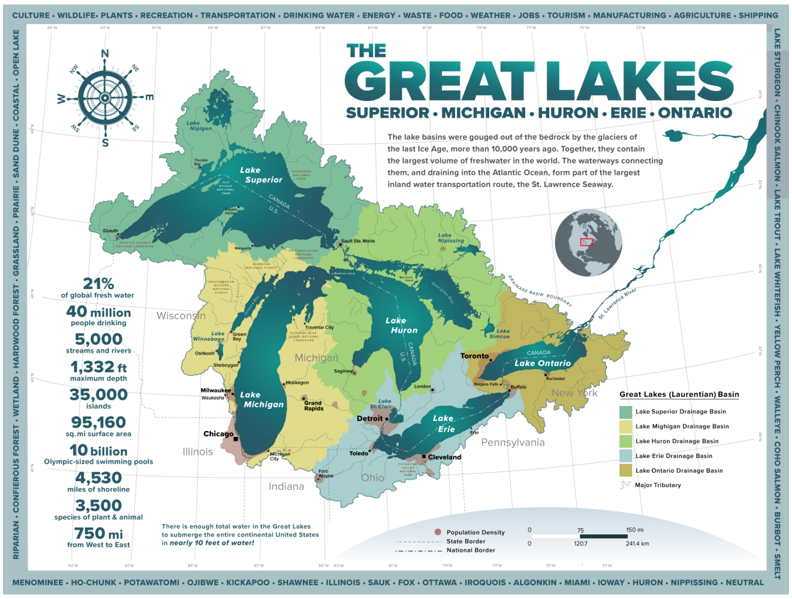


Figure 1: Introduction to the Great Lakes

In addition, the control mechanism of the Great Lakes network mainly consists of two dams: Compensating Works (Soo Locks) and Moses-Saunders Dam. However, in the face of the complex dynamic network flow of the Great Lakes, which is affected by the combined influence of many human factors and natural factors, it is sometimes difficult to adjust the two dams to take into account the entire network. Therefore, how to regulate the Great Lakes to achieve the desired goals is also a thorny issue.

1.2 Restatement of the Problem

The key to the problem is to establish a control mechanism for the Great Lakes. The specific details of the problem can be restated as:

- Establishment of the network model of the five Great Lakes.
- Determination of the optimal water levels of the five Great Lakes at any time of the year, taking into account the various stakeholders' desires.
- Develop control algorithms based on data and the network to maintain optimal water levels in the Great Lakes and use 2017 data to test the effectiveness of the algorithm.
- Analyze the sensitivity of algorithms for two dams' outflows and discuss environmental changes.

- Combined with the information given in the addendum, conduct an extensive analysis of Lake Ontario.
- Provide IJC leadership with a one-page memo describing the key features of the model to convince them to select your model.

1.3 Our Work

In order to avoid redundant descriptions and display our work process intuitively and clearly, our workflow diagram is shown in Figure 2 below:

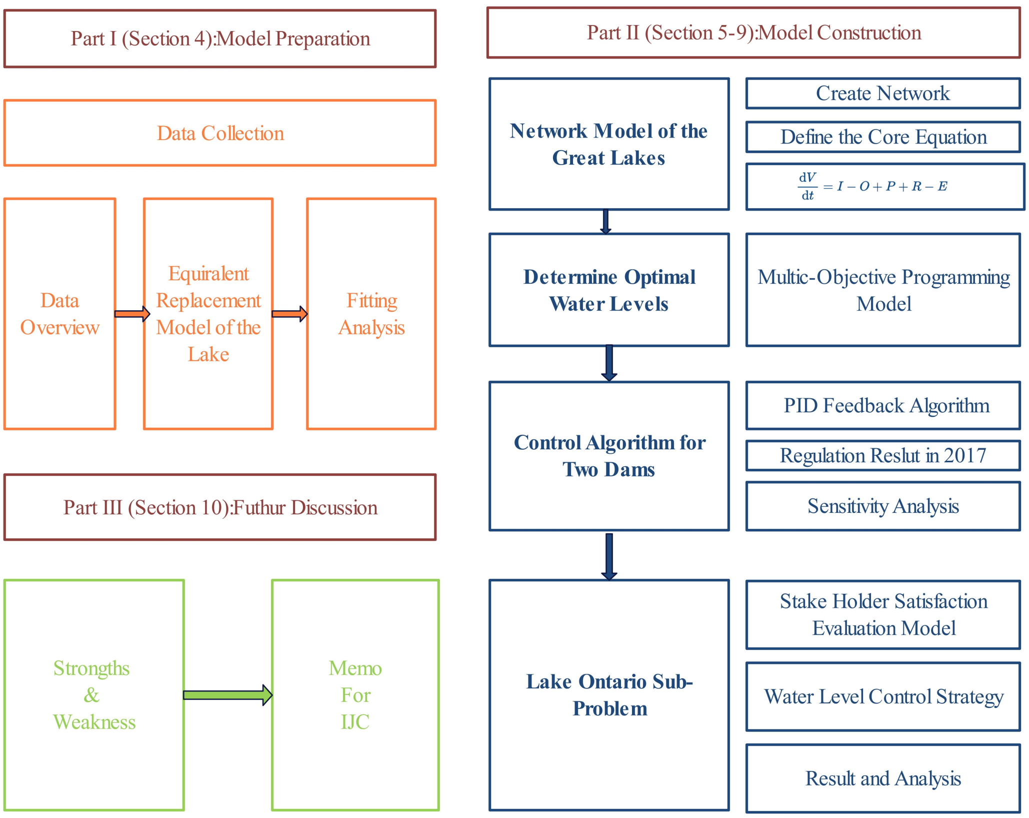


Figure 2: Flow Chart of Our Work

2 Assumptions and Explanations

Considering that practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model, and each hypothesis is closely followed by its corresponding explanation:

Assumption 1: **The evaporation and precipitation of rivers are ignored**
Explanation: The amount of precipitation and evaporation in rivers is difficult to calculate. And compared with the Great Lakes, the precipitation and evaporation of rivers are much smaller, so they will not have a significant impact on the Great Lakes network.

Assumption 2: **We don't take into account the impact of the water-taking behavior of residents along the lake**
Explanation:

Assumption 3: **The water-taking behavior of residents along the lake is uncontrollable and unpredictable. Except that, the amount of water in this part is so small. Therefore, we choose to ignore it**
*Explanation:*The historical precipitation, runoff,evaporation and other data come from authoritative websites, such as NOOA, with high accuracy.

Additional assumptions are made to simplify analysis for individual sections. These assumptions will be discussed at the appropriate locations.

3 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
i	Lake number
t	Time (calculated in months)
Z_i^t	The water level of the i-th lake in the t-th month
V_i^t	The amount of water in the i-th lake in the t-th month
P_i^t	The lake surface precipitation of the i-th lake in the t-th month
R_i^t	The runoff inflow volume of the i-th lake in the t-th month
E_i^t	The amount of water evaporated from the i-th lake in the t-th month
Q	The river flow
e	The difference between the current water level and the target water level

*There are some variables that are not listed here and will be discussed in detail in each section.

4 Model Preparation

4.1 Data preparation

4.1.1 Data collection

The problem's addendum provides the great lakes' water level data and river flow data, but it do lacks data on evaporation, precipitation, runoff, ice jams, etc. We mainly obtain these data from the NOAA official website[1].

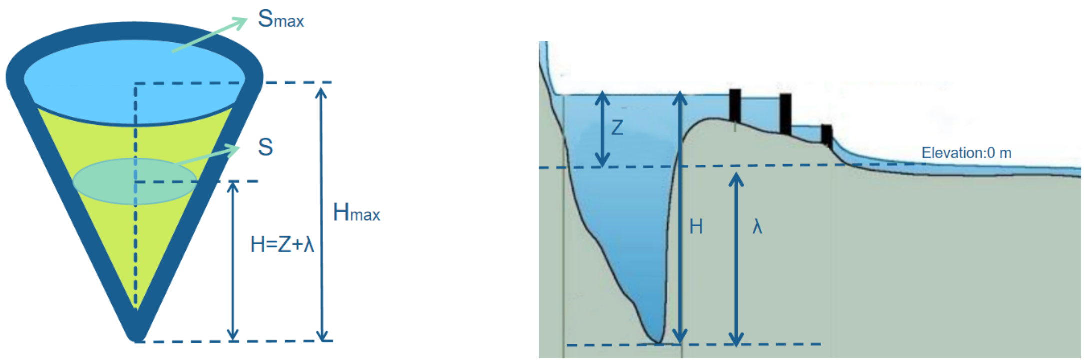
4.1.2 Data cleaning

Since the data given in the addendum is partially missing, we need to fill in the missing data appropriately. Missing data falls into two types: One is that nearly whole year's data is missing. The other type is that data for only one month in a year is missing.

For the first type of missing data, we search relevant websites to obtain data which can fill vacant positions. For the second type of missing data, we use the average value of the same month in previous years to fill in it.

4.2 Build the Equivalent Replacement Model of the Lake

To facilitate subsequent discussion, we want to determine the relationship between water level of the lake and the amount of water stored in it. Therefore we adopt an equivalent alternative approach, which is commonly used to simplify physical problems. As shown in Figure 3(a), each lake is equivalent to an inverted cone.



(a) The inverted cone model of the lake (b) Relationship between depth and water level

Figure 3: Equivalent replacement model of the lake

We assume that S_{max} represents the lake's maximum surface area and H_{max} represents the lake's maximum depth. Note the distinction between depth H and water level Z. Depth is the distance from the surface to the bottom of the lake, while water level is the distance from the water surface to the sea level. Figure 3(b) visualizes the relationship between H and Z. We convert water level to depth using formula(1).

$$H = Z + \lambda \tag{1}$$

where λ a is a constant for each river which is calculated on the basis of real data.

Analyzing the cone model, we can easily get the proportional relationship between the lake's current surface area S and the maximum surface area S_{max}.

$$\frac{S}{S_{max}} = (\frac{Z+\lambda}{H_{max}})^2 \tag{2}$$

Since we have obtained S and H, we can calculate the amount of water stored in the lake V by applying Vertebral volume formula.

$$V = \frac{1}{3}SH = \frac{(Z+\lambda)^3 S_{max}}{3H_{max}^2} \tag{3}$$

We collect the real $H_{max}, S_{max}, \lambda$ data of each lake and present them in Table 2.

Table 2: The collected data for $H_{max}, S_{max}, \lambda$

	Superior	HUR-MIC	St.Clair	Erie	Ontario
$S_{max}(km^2)$	82100	117400	1100	25670	19010
$H_{max}(m)$	140.8	68.8	8.2	18.0	82.4
$\lambda(m)$	-42.1	-107.2	-171.0	-154.9	8.4

4.3 Fitting Analysis

4.3.1 Precipitation, Runoff and Evaporation

After analyzing and studying the historical data of Precipitation, Runoff and Evaporation in the great lakes basin, we conclude that these factors are closely related to time and have obvious monthly patterns. Therefore, we perform polynomial fitting analysis on the three factors and time. The polynomial we use is shown in formula(4).

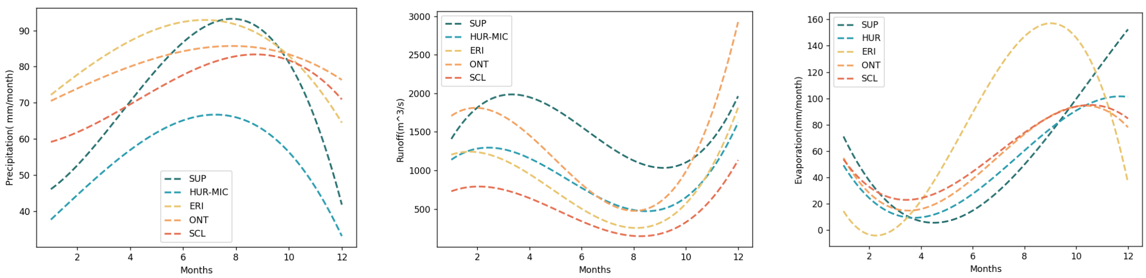
$$\hat{y} = a_0 t^n + a_1 t^{n-1} + a_2 t^{n-2} + ... + a_{n-1} t + a_n \tag{4}$$

where $\hat{y}$ represents predictive value of a specific factor; t represents time which takes month as unit.

The goal of the fitting process is to minimize the sum of squares of residuals:

$$MIN \left\{ \varepsilon = \sum_{i=1}^{12} (\hat{y}_i - y_i)^2 \right\} \tag{5}$$

Finally, we obtain each lake's fitting functions of Precipitation, Runoff and Evaporation. Based on the results, we draw the curves shown in Figure 4. These fitting functions will play an important role in later sections.



(a)Precipitation (b)Runoff (c)Evaporation

Figure 4: The fitting curves of Precipitation, Runoff and Evaporation in each lake's basin

4.3.2 River Flow

Rivers connect neighboring lakes and is seen as a essential part of the great lakes system. Research shows that river flow is related to the water levels of upstream and downstream lakes. FRANK H. QUINN gives the following formula to calculate river flow[2]:

$$Q = k(Z_{up} - b)^2 (Z_{up} - Z_{down})^{0.5} \tag{6}$$

where Q represents the river flow; Z_{up} and Z_{down} represent the water levels of the upstream and downstream lakes respectively; k and b are constants depend on the specific lake.

To obtain k and b in formula(6), we use the data provided in the Problem Addendum, which includes information about water levels and river flows. We performed fitting analysis trying to find the (k, b) combination that minimizes the ε defined in Formula(5). This process is taken for every river in the great lakes system. The k and b values of each river are finally determined. Table 3 presents the results of our work.

Table 3: The k and b values of each river

	St.Mary	St.Clair	Detroit	Niagara	St.Lawrence
k	3.24	2.24e-6	77.98	13.32	1.48
b	167.36	48421.77	165.65	167.58	50.04

5 Network Model of the Great Lakes

Based on the map of the Great Lakes[3], we established a directed network graph with lakes as nodes and rivers as edges. The direction of each edge is from the upstream lake to the downstream lake. According to the actual hydrological conditions and relevant literature, we treat Lake Huron and Lake Michigan as a whole named Lake Huron-Michigan. In addition to the five Great Lakes, we also consider Lake St. Clair between Lake Huron and Lake Erie as a node. The preliminary network is shown in Figure 5.

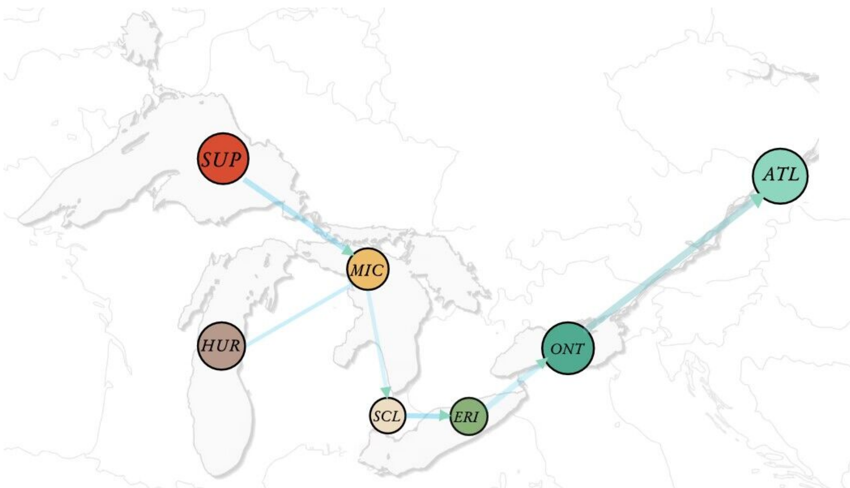


Figure 5: Great Lakes Network Map

With the preliminary network map in hand, we need to further calculate the water levels of each lake. According to the National Oceanic and Atmospheric Administration (NOAA) survey of the Great Lakes[4], the water volume V of the Great Lakes mainly depend on the river inflow I, the river outflow O, and the precipitation P, evaporation E and runoff from the lake's land drainage area R. Figure 6 visualizes these factors. The relationship between the above variables can be described by the **water conservation equation** shown in formula(7).

$$\frac{\mathrm{d}V}{\mathrm{d}t} = I - O + P + R - E \tag{7}$$

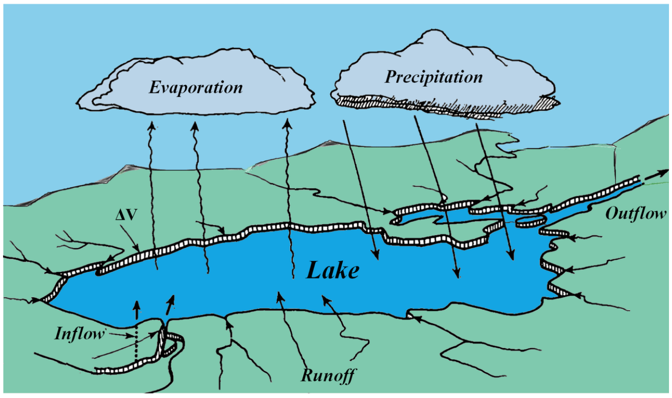


Figure 6: The source and destination of lake water

Since our data is monthly measured, formula (7) needs to be discretized.

$$V^{t+1} - V^t = I^t - O^t + P^t + R^t - E^t \tag{8}$$

where t is the month; In addition, the understanding of this formula is the monthly conservation of water.

Returning to the Great Lakes network, we **use i to represent lakes**, i from 1 to 5 representing Lake Superior, Lake Huron-Michigan, Lake St. Clair, Lake Erie and Lake Ontario respectively. Therefore, for the node $Lake_i$ in the network model, we have:

$$V_i^{t+1} - V_i^t = I_i^t - O_i^t + P_i^t + R_i^t - E_i^t \tag{9}$$

Adjacent lake nodes are related through rivers, from which we can get formula (10):

$$O_i^t = I_{i+1}^t = Q_i^t \times T \tag{10}$$

Where O_i^t is the outflow of the i-th lake, I_{i+1}^t is the inflow of the i+1 lake, Q_i^t is the flow of the river flowing out of the i-th lake, and T is the time of one month, which is 2678400 seconds.

In the model preparation part, through historical data fitting, we have obtained the expected monthly precipitation P, evaporation E, and runoff R for each lake. We also establish the cone model to describe the relationship between the lake's water volume V and the water level Z, as well as the functional fitting relationship between the river flow and the water levels of the upstream and downstream lakes. Therefore, we can relate the different lakes' water level and integrate the network by rivers. Figure 7 shows the complete network model of the great lakes.

6 Determination of the Optimal Water Levels

6.1 The Establishment of Multi-objective Programming Model

6.1.1 Analyze stakeholders

Our goal is to adjust the water levels of the Great Lakes to ensure the greatest overall benefit to all stakeholders, so we first discuss the stakeholders. Based on the question prompts and querying relevant information[5], we summarized the relevant stakeholders into four types:

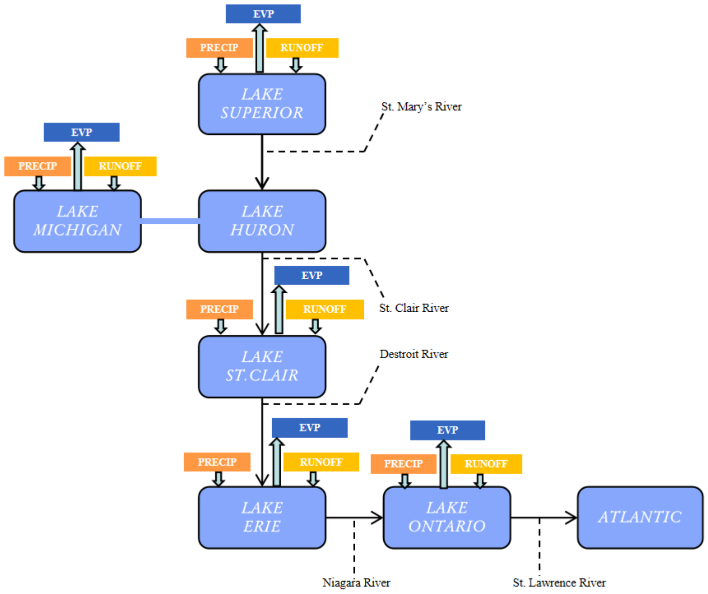


Figure 7: Great Lakes Network Model

1) **Coastal residents**: Coastal residents want moderately stable water levels. If the water level is too high, it may cause flooding and cause huge property damage.

2) **Shipping companies**: Shipping companies want higher water levels to ensure unimpeded passage of ships, so they have requirements for minimum water levels.

3) **Power generation companies**: Power generation companies want to maintain higher water levels to provide better power production efficiency. So they also have minimum power generation water level requirements.

4) **Environmentalists**: Environmentalists want stable and fluctuating water levels as this is better for maintaining ecosystems.

These four types of stakeholders have different expected water levels, as well as different maximum or minimum acceptable water levels. Involving a lot of symbols, we have organized them into the following Table 4:

Table 4: Symbol description table

Stakeholder type	Coastal residents	Shipping companies	Power generation companies	Environmentalists
Ideal water level	$Ideh_i^{(resident)}$	$Ideh_i^{(ship)}$	$Ideh_i^{(power)}$	$Ideh_i^{(ecosys)}$
Minimal water level	$None$	$Minh_i^{(ship)}$	$Minh_i^{(power)}$	$Minh_i^{(ecosys)}$
Maximal water level	$Maxh_i^{(resident)}$	$None$	$None$	$Maxh_i^{(ecosys)}$

In Table 4, $Ideh_i$ represents the optimal water level that stakeholder i wants, $Maxh_i$ represents the highest water level that stakeholder i can accept, and $Minh_i$ represents the lowest water level that stakeholder i can accept. $None$ means that no upper or lower limit for acceptable water levels has been set.

As for the values of the above water levels, we determined the minimum or maximum water levels that different stakeholders can tolerate by investigating the data on the International Joint Commission website[6]. In addition, we have held discussions, meetings and articles to identify ideal water levels for various stakeholders.

6.1.2 Objective function

- **Target1: Lake water levels meet the expectations of all stakeholders**

Different types of stakeholders often have conflicts over water levels, making it difficult to meet their expectations and goals at the same time. So, we measure the degree of realization of the goals of these stakeholders by calculating the weighted sum of the deviations between the actual water level and each stakeholder's ideal water level. So we give the following objective function:

$$\begin{aligned} Benefit_i = \alpha \sum_{t=1}^{12} (Z_i^t - Ideh_i^{(resident)})^2 + \beta \sum_{t=1}^{12} (Z_i^t - Ideh_i^{(ship)})^2 \\ + \gamma \sum_{t=1}^{12} (Z_i^t - Ideh_i^{(power)})^2 + \delta \sum_{t=1}^{12} (Z_i^t - Ideh_i^{(ecosys)})^2 \end{aligned} \tag{11}$$

where $Benefit_i$ represents the first objective function. α,β,γ and δ are the weights of the four types of stakeholders.

- **Target2: Lake water level remains stable**

We do not want the lake level to change too much, as this may affect the interests of many people. Ideally lake levels should be stable. Stability can be measured in terms of range and variance. To do this, we calculate the difference between the lake level and the mean water level, and the range of the lake level to express objective 2. Since the values of the range and variance may be small, we use exponential functions to amplify them. The following objective function is obtained:

$$Stability_i = e^{\theta Range_i + \varphi Variance_i} \tag{12}$$

Where $Stability_i$ is the second objective function, $Range_i$ is the range, and $Variance_i$ is the standard deviation.Their calculation formula is as follows:

$$Range_i = max\left\{Z_i^t\right\} - min\left\{Z_i^t\right\} \tag{13}$$

$$Variance_i = \frac{1}{12}\sum_{t=1}^{12}(Z =_i^t - \frac{1}{12}\sum_{t=1}^{12} Z_i^t)^2 \tag{14}$$

6.1.3 Restrictions

- **Water level constraint**

The water level of each lake should meet the minimum and maximum water level requirements of the four types of stakeholders. Therefore, the water level constraint is as follows:

$$\text{MAX}\left\{Minh_i^{ship}, Minh_i^{power}, Minh_i^{ecosys}\right\} < Z_i^t < MIN\left\{Maxh_i^{resident}, Maxh_i^{ecosys}\right\} \tag{15}$$

- **Water volume constraint**

The change in the amount of water in each lake is equal to the difference between the amount of water entering the lake and the amount of water leaving the lake. We have already obtained this relationship when building the network earlier, as shown in the following formula.

$$V_i^{t+1} - V_i^t = I_i^t - O_i^t + P_i^t + R_i^t - E_i^t \tag{16}$$

- **Lake model constraints**

There is a relationship between the water volume and water level of the lake. We have established an equivalent model to describe the relationship between them in Section 4.2. The constraint formula is as follows:

$$V_i^t = \frac{(Z_i^t + \lambda)^3 S_{max}}{3H_{max}^2} \tag{17}$$

6.1.4 Multi-objective programming model

By summarizing the above objective functions and constraint equations, we can obtain the following multi-objective programming model for determining the monthly optimal water levels in the Great Lakes.

$$MAX\left\{Benefit_i, Stability_i\right\} \quad i = 1, 2, 3, 4, 5 \tag{18}$$

$$s.t.\begin{cases} Z_i^t < MIN\left\{Maxh_i^{(resident)}, Maxh_i^{(ecosys)}\right\} \\ Z_i^t > MAX\left\{Minh_i^{(ship)}, Minh_i^{(power)}, Minh_i^{(ecosys)}\right\} \\ V_i^{t+1} - V_i^t = I_i^t - O_i^t + P_i^t + R_i^t - E_i^t \\ V_i^t = \frac{(Z_i^t+\lambda)^3 S_{max}}{3H_{max}^2} \end{cases} \tag{19}$$

6.2 Parameter determination

- **Target1:** α,β,γ,δ

Combining the relevant data queried[8], from the perspective of the public, we hope that the interests of coastal residents will be protected with the highest priority, followed by the interests of shipping companies and power generation companies, and finally ecological interests will be considered. The Target1's parameter values are shown in Table 5 below.

Table 5: Parameter values of Traget1

Parameter	α	β	γ	δ
Value	0.3	0.25	0.25	0.2

- **Target2:** θ,φ

From a statistical point of view, the range reflects the information of less data, while the variance reflects the information of all data. The information content of the variance is higher than the range, so a higher weight is set for the variance and a higher weight for the range. low weight. The Target2's parameter values are shown in Table 6 below.

Table 6: Parameter values of Traget2

Parameter	θ	φ
Value	0.3	0.7

6.3 Optimal water level calculation results

We used the goal programming solver provided by Lingo language to solve this multi-objective programming model, and obtained the optimal water levels for each month of the year in the Great Lakes. The water level values are shown in Table 7

Table 7: Optimal water levels

Lake	Superior	HUR-MIC	St.Clair	Erie	Ontario
Jan	183.96	176.58	175.41	175.34	74.80
Feb	182.80	176.20	176.09	173.98	75.05
Mar	182.95	175.88	175.05	174.27	75.19
Apr	182.76	177.91	175.35	174.98	75.76
May	183.33	175.16	174.77	174.77	74.83
Jun	183.10	175.86	175.20	174.09	75.56
Jul	183.90	175.67	175.29	174.77	74.93
Aug	182.09	177.21	175.55	174.58	75.04
Sep	183.23	177.02	176.61	174.59	75.57
Oct	183.24	176.78	175.55	174.65	75.16
Nov	183.22	177.42	175.13	174.87	75.40
Dec	183.61	176.09	175.51	173.65	75.10

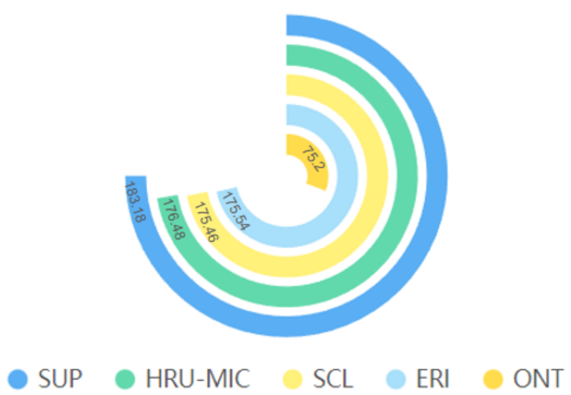


Figure 8: Annual optimal water levels

Although there is an optimal water level every month, it is difficult to actually control it, so we averaged the optimal water levels in 12 months as the optimal water level for a year (the results are shown in Figure 8). Later models will also use the annual optimal water level as the target to calculate and solve.

7 Control algorithm for two dams

7.1 Establishment the Control algorithm

In Section 6, we calculated the monthly optimal water levels and the annual average optimal water levels for the five lakes. However, water levels fluctuate under natural situation. To be able to maintain optimal water levels, we need to take fully advantage of two control mechanisms along the water flow path: Compensating Works(Soo Locks) and the Moses-Saunders Dam. Therefore, we propose a **PID feedback** control algorithm[7]. The core idea of this algorithm is to adjust based on the error between the current measured water level and the optimal water level to reduce the error at the next measurement time point. The details of the PID feedback control algorithm are as follows.

Firstly, we consider the time interval and target water level of regulation. There are two feasible options: One is to take the monthly optimal water level as target and perform adjustments once a day. The other is to take the annual average optimal water level as target and regulate once a month. In the end, we choose the second option because it take into account the cost of running the dams and the time required for regulation to take effect.

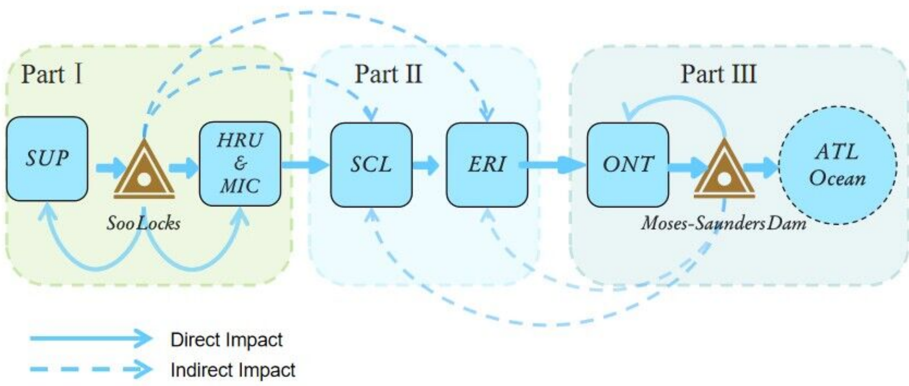


Figure 9: The three parts of the great lakes system

Secondly, we divide the great lakes system into three parts based on the control scope of the two dams. As shown in Figure 9, Part I consists of Lake Superior and Lake Huron-Michigan, which are directly affected by the Soo Locks. Lake St. Clair and Lake Erie make up the Part II. These two lakes are special in that the Soo Locks and Moses-Sanders Dam cannot directly control the river flows upstream or downstream of them. In other words, they are affected by the two dams indirectly. Part III is Lake Ontario, whose outflow is directly controlled by the Moses-Sanders Dam. Due to the vast distance, the Soo Locks' regulatory impact on Lake Ontario is almost negligible compared to that of the Moses-Saunders Dam. Likewise, we do not consider the impact of Moses-Sanders Dam on Lake Superior and Lake Huron-Michigan.

After the above works, we can perform the PID feedback algorithm. As shown in formula(20), the general style of this algorithm includes three parts:Portion, Integer and Derivation(from first item to last item). The output of the Portion Part is proportional to the input error. If only the Portion Part is used, the system will have a steady-state error, and an Integer Part needs to be added. The role of the Derivation Part is to predict the changing trend of the error and provide measures to suppress the error change in advance. The final output of the PID algorithm is a linear combination of the three parts.

$$RF(t) = k_p e(t) + k_i \int_0^t e(t)\mathrm{d}t + k_d \frac{\mathrm{d}e(t)}{\mathrm{d}t} \tag{20}$$

where $RF(t)$ represents the output of the algorithm which is also called the regulatory factor because we use it to regulate two dams' outflow; $e(t)$ represents the water level error at time t; k_p, k_i, k_d is combination coefficients.

Since we measure the water level once a month, the formula(20) needs to be discretized.

$$RF^t = k_p e^t + k_i \sum_{j=1}^{t} e^j + k_d(e^t - e^{t-1}) \tag{21}$$

Based on the above formula, the specific steps to regulate two dams' outflow are as follows.

Step1: Calculate the water level error of each lake

Water level error is the difference between the lake's current water lever and optimal water level. We calculate the water level error for each of the five lakes using formula(22).

$$e_i^t = Z_i^t - Z_i^{opt} \tag{22}$$

where e_i^t represents the water level error of the i-th lake in the t-th month; Z_i^t represents the water level of the i-th lake in the t-th month, while Z_i^{opt} represents the annual average optimal water level of i-th lake.

Step2: Calculate total error of each dam

The total error of the dam depends on the water level errors of the lakes which are affected directly or indirectly by the dam's regulation. Taking the Soo Locks as an example, the dam directly controls Lake Superior and Lake Huron-Michigan, indirectly affects Lake St.Clair and Lake Erie. Therefore the total error of Soo Locks should be the weighted sum of these four lakes' water level error. When determining the weight of each lake's error, we comprehensively consider factors such as the distance between the lake and the dam, the maximum capacity of the lake, and the dam's ability to control the lake. Finally we obtain the formula(23) to calculate the Soo Locks' total error.

$$te_s^t = 0.7e_1^t + 0.2e_2^t + 0.015e_3^t + 0.085e_4^t \tag{23}$$

where te_s^t represents the total error of Soo Locks in t-th month; $e_1^t, e_2^t, e_3^t, e_4^t$ respectively represent the water level errors in t-th month of Lake Superior, Lake Huron-Michigan, Lake St.Clair, and Lake Erie.

Similarly, we can calculate the total error of Moses-Saunders Dam, which is the weighted sum of the water level errors of Lake Ontario, Lake St.Clair, and Lake Erie.

$$te_m^t = 0.05e_3^t + 0.1e_4^t + 0.85e_5^t \tag{24}$$

where te_m^t represents the total error of Moses-Saunders Dam in t-th month; e_3^t, e_4^t, e_5 respectively represent the water level errors in t-th month of Lake St.Clair, Lake Erie and Lake Ontario.

Step3: Calculate regulatory factor

Regulatory factor is the output of PID equation. On the basis of formula(21), we define the PID equations of Soo Locks and Moses-Saunders Dam.

$$RF_s^t = k_{ps} \cdot te_s^t + k_{is} \sum_{j=1}^{t} te_s^j + k_{ds} \cdot (te_s^t - te_s^{t-1}) \tag{25}$$

$$RF_m^t = k_{pm} \cdot te_m^t + k_{im} \sum_{j=1}^{t} te_m^j + k_{dm} \cdot (te_m^t - te_m^{t-1}) \tag{26}$$

where RF_s^t, RF_m^t respectively represent the regulatory factors of Soo Locks and Moses-Sanders Dam in t-th month; k_{ps}, k_{is}, k_{ds} and k_{pm}, k_{im}, k_{dm} are combination coefficients of two dams' PID equation. The method to determine combination coefficients will be discussed in the sensitivity analysis of the control algorithm.

Step4: Perform regulation of two dams' outflow

We use the regulatory factors obtained in step3 to regulate the outflow of the two dams. The result is the changes in St. Mary's river flow and St.Lawrence river flow. The new flow calculation formulas for these two rivers are shown in the following formulas.

$$Q_1^{t\prime} = Q_1^t + RF_s^t \tag{27}$$

$$Q_5^{t\prime} = Q_5^t + RF_m^t \tag{28}$$

where $Q_1^{t\prime}, Q_5^{t\prime}$ represents the regulated flow in t-th month, while Q_1^t, Q_5^t represents the original flow in t-th month.

After the above four steps, we can get river flow after regulation. By plugging the new flow into the network model, the water levels in the next month can be obtained. These water levels will be closer to the annual average optimal water levels. Then we recalculate the water level errors in the next month and repeat steps 1-4. As time goes by, the water level errors gradually decreases, and the water levels eventually stabilize at the optimal water level. We draw a flowchart of the regulatory process, as shown in Figure 10.

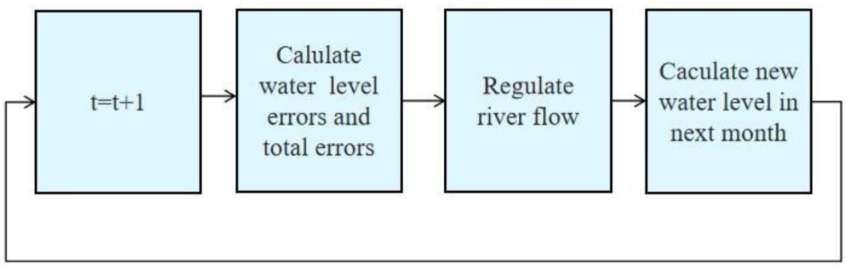
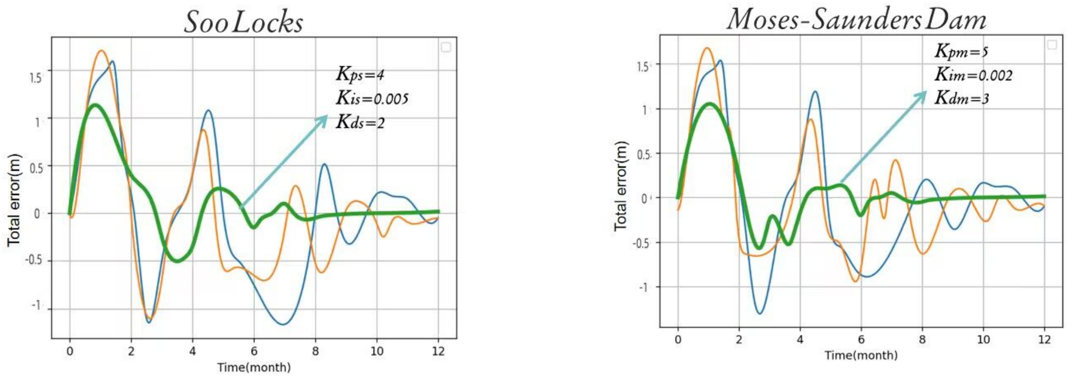


Figure 10: Changes in prioritization pyramid

7.2 The sensitivity of the control algorithm

We measure the sensitivity of the control algorithm using the fluctuation degree of the algorithm's error regulation process and the time it takes to reach new steady state. The combination coefficients k_{ps}, k_{is}, k_{ds} and k_{pm}, k_{im}, k_{dm} of the two dams' PID equations play important roles in determining the algorithm's sensitivity. Reasonable combination coefficients values help to reduce fluctuations in the regulation process and speed up the arrival of steady state.

The selection of correlation coefficients is often based on experience and trial. We collect relevant literature and historical data, initially obtain the feasible range of these coefficients, then compare the sensitivity of the model under different coefficient combinations, and finally determine the optimal combination coefficients values.



(a) Soo Locks (b) Moses-Saunders Dam

Figure 11: The two dams regulation process curves

As shown in Figure 11, we set water level errors of great lakes which cause the total errors of two dams at time 0. Then we simulate the regulation process of the two dams. Different color curves in Figure 9 use different combination coefficients. It is found that the green curves show the best regulation effect. And their coefficients are what we need. The optimal k_{ps}, k_{is}, k_{ds} of Soo Locks are 4, 0.005, 2, while the optimal k_{pm}, k_{im}, k_{dm} of Moses-Saunders Dam are 5, 0.002, 3. In this case, we can say that the algorithm has very good sensitivity.

7.3 Regulation results of 2017

We simulate the regulation process of the control algorithm on the basis of data in 2017. The five great lakes' water levels in each month after the regulation are calculated. We visualize calculated water levels, original recorded water levels and annual average optimal water levels for five lakes in **Figure 12**. It is obvious that the water levels after regulation are closer to the optimal water levels than the original recorded water levels. So we can reasonably infer that various stakeholders would be more satisfied

8 Changes in Environmental Conditions

Environmental conditions are the essential part of the Great Lakes' network. In the previous discussion, we use historical data to perform polynomial fit analysis on environmental conditions such as precipitation, runoff, and evaporation to obtain their relationship with time. In other words, we take full account of seasonal changes in these environmental conditions. But climate change may lead to some abnormal changes in these conditions. Therefore we will discuss the sensitivity of our algorithm to abnormal changes in precipitation, winter snowpack, and ice jams respectively.

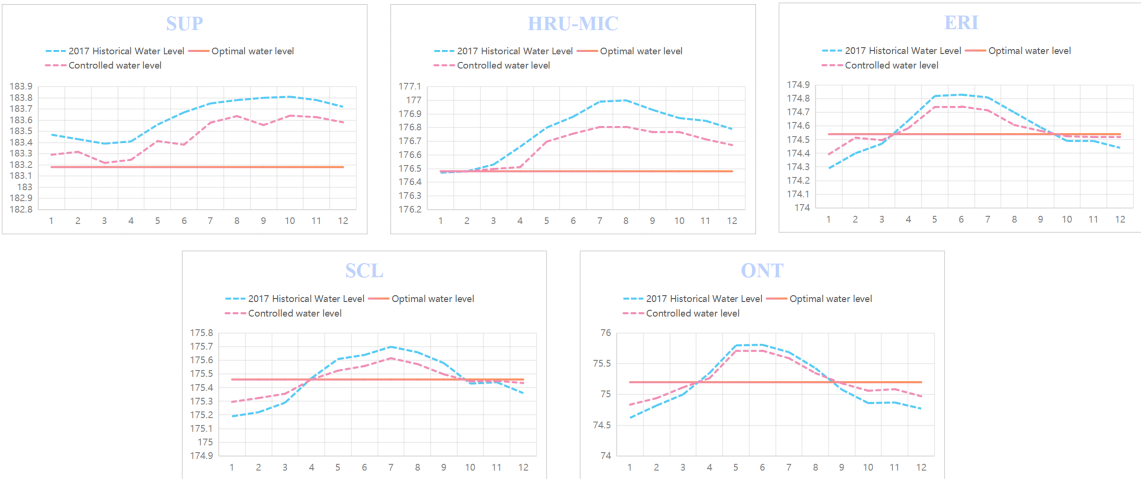


Figure 12: The three types of water levels in 2017

8.1 Precipitation Changes

Our algorithm is robust to seasonal changes in precipitation. Therefore, we only need to study the impact of abnormal precipitation changes on our control algorithm. We let the precipitation deviate from the original amount $+30\%, -30\%, +60\%, -60\%$ respectively, then perform the control algorithm, and finally calculate the degree of the Great Lakes water level deviating from the optimal water level after one year's regulation.

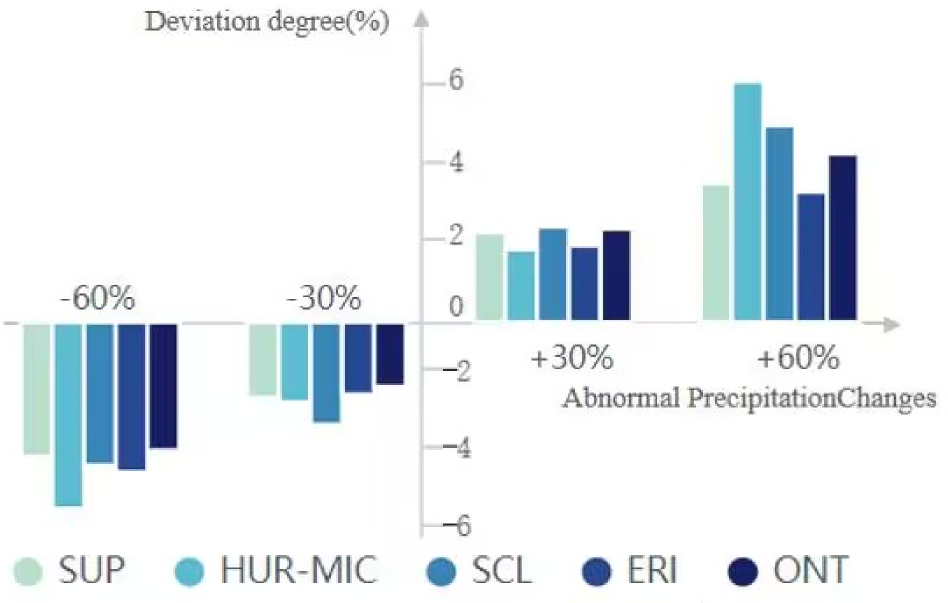


Figure 13: Deviation of water levels caused by precipitation changes

Figure 13 visualizes the calculation result. And we can conclude that our control algorithm can alleviate abnormal precipitation changes to a certain extent, but this regulation ability is limited and cannot handle excessive precipitation changes.

8.2 Winter Snowpack Changes

Winter snowpack mainly affects runoff in the lakes' basin. Figure 4(b) in section 4.3.1 shows the seasonal changes of runoff throughout the year. We can clearly see that runoff reaches a peak in spring, which is largely due to the massive melting of snowpack. As in the previous section, we simulate the abnormal winter snowpack changes, and calculate the deviation degree of the water level after one year's regulation. As shown in Figure 14, our control algorithm can alleviate the abnormal changes of winter snowpack to a certain extent.

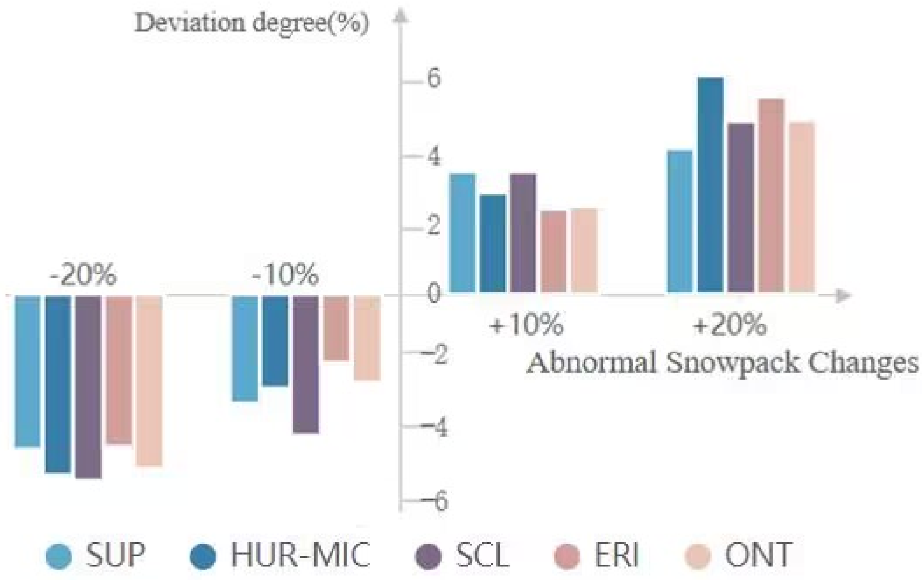


Figure 14: Deviation of water levels caused by snowpack changes

8.3 Ice jams changes

Ice jams is a factor that we have ignored before. Increased degree of ice jams will affect river flow and obstruct vessel traffic. Severe ice jams may even cause flooding. The impact of ice jams can be measured by the ice retardation rate, and formula (29) shows the new method for calculating river flow taking ice jams into account.

$$Q_i^t = k_i(Z_i^t - b_i)^2(Z_i^t - Z_{i+1}^t)^{0.5} - QR_i^t \tag{29}$$

where QR_i^t represents the ice retardation rate of lake i in the t-th month; The rest of formula (24) is consistent with what we used in section 4.3.2 formula(6).

We study the abnormal changes of ice jams like the previous two parts. Figure 15 shows the calculated deviation degree caused by ice jams changes. Since our previous model do not consider ice jams, the control algorithm is more sensitive to ice jams changes than other factors.

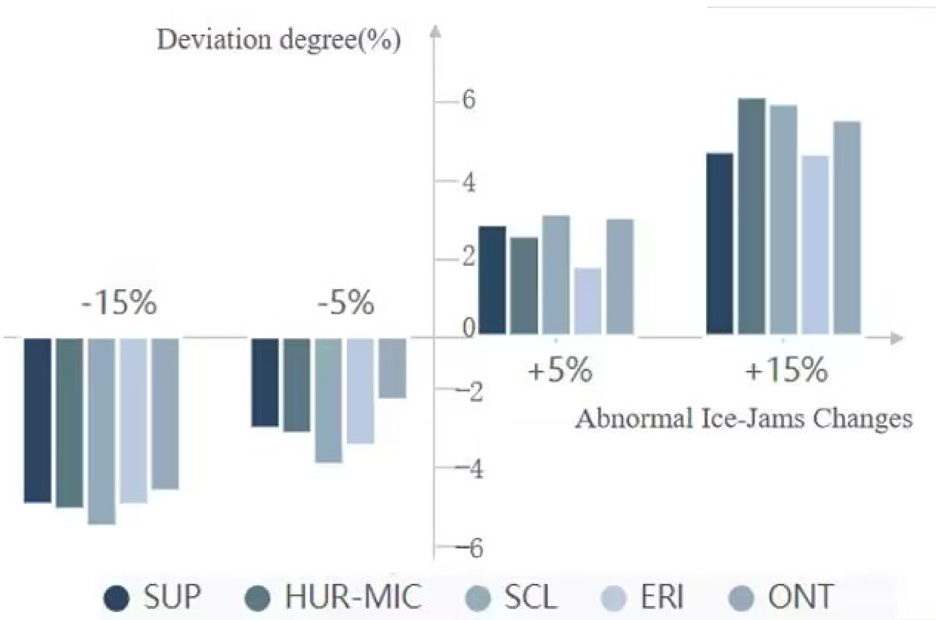


Figure 15: Deviation of water levels caused by ice jams

9 Discussion about Lake Ontario Sub-Problem

9.1 Lake Ontario stakeholder satisfaction evaluation model

Only Lake Ontario stakeholders need to be considered in the Lake Ontario sub-problem. According to the information given in the attachment to the question, the main stakeholders are: **shipping companies**, **people who manage shipping docks or live near Montreal harbor**, **environmentalists**, **property owners on the shores of Lake Ontario**, **recreational boaters and fishing boats on Lake Ontario**, and **hydro-power generation company**. In order of mention, they are represented by j=1,2,3,4,5,6 respectively. In order to measure the satisfaction of various stakeholders with water levels, **a satisfaction evaluation model** is proposed. The specific calculation steps are divided into three steps:

- Step1: Determine the expected water level range of each stakeholder.
- Step2: Calculate the satisfaction value B_j of each relevant stakeholder based on one year's water level data.
- Step3: Draw a radar chart to reflect the satisfaction of various stakeholders with the water level of the year.

For step1, the water level of a river is positively related to its flow, so the flow can be used as an equivalent substitute for the water level. After searching for relevant information, we obtained the range of water level or flow expected by each stakeholder, as shown in Table 8.

Table 8: Expected water level and flow range

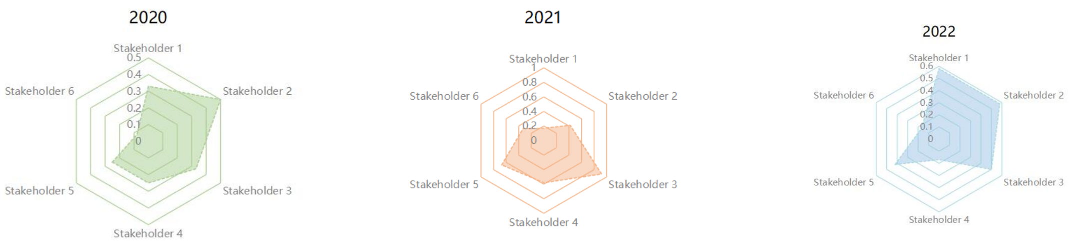
Stakeholders(j)	**1**	**2**	**3**	**4**	**5**	**6**
Object	St. Lawrence River	St. Lawrence River	Lake Ontario	Lake Ontario	Lake Ontario	Lake Ontario
Unit	m^3/s	m^3/s	m	m	m	m
Expected range	[8500,9000]	[7500,8000]	[74.5,75]	[74.4,74.8]	[74.4,74.8]	[74.8,75.5]

For step2, calculate the average water level for each month of the year. If the average water level of the month is within the expectation range of a certain stakeholder, the stakeholder's satisfaction level will be +1. Otherwise, the stakeholder's satisfaction level will be -1. Let $n_j^{satisfied}$ represents the number of months in a year when the j-th stakeholder is satisfied with the water level, and $n_j^{unsatisfied}$ represents the number of months in a year when the j-th stakeholder is not satisfied with the water level. Then **the satisfaction value** B_j of the j-th stakeholder in that year can be calculated by the following formula (30):

$$B_j = \frac{n_j^{satisfied} - n_j^{unsatisfied} + 12}{24} \tag{30}$$

In the formula, B_j is added to 12 and divided by 24 to control the value range of B_j within [0, 1]. The closer the value is to 1, the higher the degree of satisfaction, and the closer the value is to 0, the lower the degree of satisfaction.

For step3, We substituted the data from 2020 to 2022, calculated the satisfaction value of each relevant stakeholder, and expressed it with a radar chart (Figure 16).



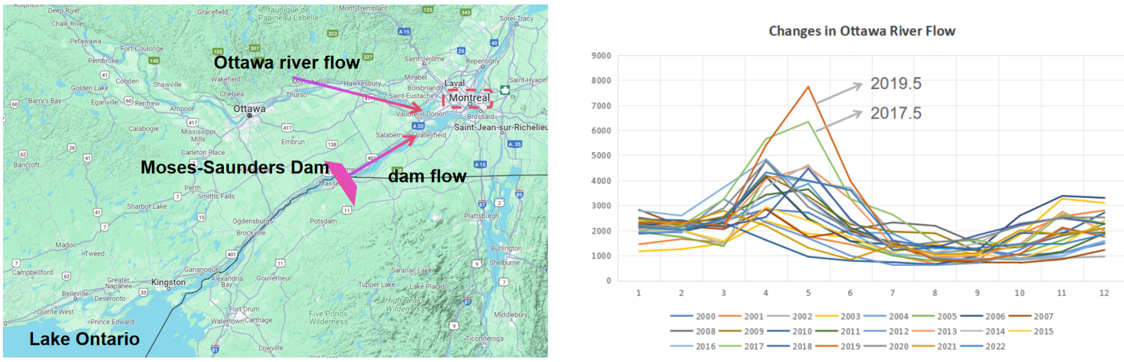
(a)2020 (b)2021 (c)2022

Figure 16: Stakeholder satisfaction radar chart

It can be seen from the radar chart(Figure 16) that the satisfaction levels of the same stakeholders in different years are not necessarily the same, and power generation companies are always not satisfied with the existing water level, and the water level strategy needs to be improved.

9.2 Lake Ontario water level control strategy

The flow of the St. Lawrence River is mainly determined by the flow of the Moses Sanders Dam on the St[9]. Lawrence River and the flow of the Ottawa River that merges into the St. Lawrence River, as shown in Figure 17(a). And you can see from the picture that the city of Montreal is located at the intersection of the two, and is most directly and obviously affected by the flow of the St. Lawrence River. To prevent dangerous surges caused by excessive flows in the St. Lawrence River, a balance needs to be maintained between the flow from the dam and the flow from the Ottawa River. From Figure 17(b), we can see that the flow of the Ottawa River is larger in April, May, and June, and smaller in August, September, and October. Correspondingly, the dam flow should be adjusted smaller in April, May, and June, and the dam flow should be adjusted larger in August, September, and October. Considering the impact of snow on the flow of the Ottawa River, extreme flows like those in May 2017 and May 2019 in Figure 17(b) may occur, so it can be increased in January and February when the flow of the Ottawa River is not large. Dam flow, lower the Lake Ontario water level in advance to prevent the Lake Ontario water level from rising beyond the threshold due to reduced dam flow in April, May, and June.



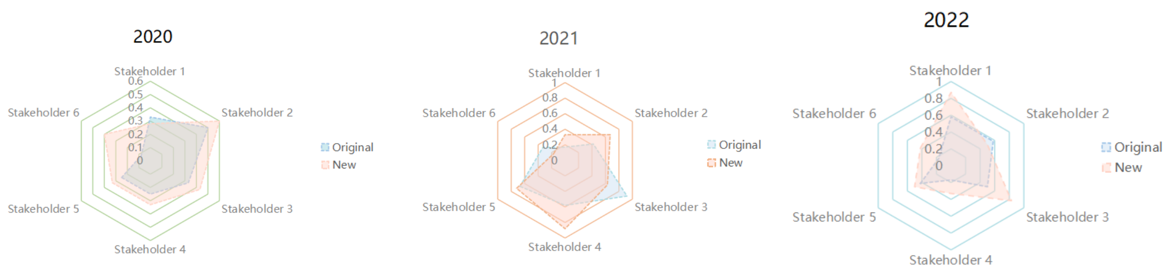
(a) St.lawrence river map (b) Ottawa River flow over the years

Figure 17: Basis for adjustment plan

The impact of the regulation strategy on water level and flow can be simply quantified: the dam flow increased slightly in January and February, the corresponding monthly average water level data of Lake Ontario decreased by 0.05 m, and the corresponding monthly flow of the St. Lawrence River increased by 100 m^3/s. The dam flow decreased in April, May and June, the corresponding monthly average water level data of Lake Ontario each increased by 0.1 m, and the corresponding monthly flow of the St. Lawrence River decreased by 200 m^3/s. In August, September and October, the dam flow increased, the corresponding monthly average water level data of Lake Ontario each decreased by 0.1 m, and the corresponding monthly flow of the St. Lawrence River increased by 200 m^3/s. The remaining months remain unchanged.

9.3 Results and analysis

We used the strategy in Section 9.2 to adjust the data for the three years from 2020 to 2022, and used the adjusted data to calculate the annual satisfaction value of each stakeholder, and represented it with a new radar chart (Figure 18).



(a)2020 (b)2021 (c)2022

Figure 18: Radar chart of stakeholder satisfaction before and after implementation of adjustments

In the radar chart in Figure 18, the area after adjustment (pink part) is significantly larger than the area before adjustment (blue part). Although the satisfaction level of some relevant stakeholders will decrease after the adjustment, the overall satisfaction level will be significantly improved. It shows that our plan is reasonable.

10 Strengths and Weaknesses

10.1 Strengths

- **Comprehensive consideration**
  We model Great Lakes water levels by taking into account evaporation, rainfall, runoff, and inflows and outflows from major rivers. Changes in rainfall, snow cover, and river ice jams were also discussed. Make the model more comprehensive.
- **Have authenticity**
  Our article focuses on the actual geography and hydrology of the Great Lakes and uses extensive use of real lake and river data for modeling and solution. It can better reflect the real situation of the Great Lakes.
- **High degree of visualization**
  We have made some visual displays of the results of the main questions, and the visualization types are very diverse, including line charts, bar charts, radar charts, etc. Easy for readers to read and understand.
- **Good expandability**
  Our model can be extended to larger and more complex lake systems, model new lake system networks, and combine the network with the PID feedback adjustment mechanism to regulate the entire lake system.

10.2 Weaknesses

- **Rough approximation and prediction**
  For lake precipitation, evaporation, runoff and other conditions, no time series has been established to accurately predict, but a value is simply fitted based on previous years' data and substituted into the model calculation.
- **Model parameters are not optimal**
  Some parameters in the model we established did not further explore their optimal values, but only adopted a reasonable value and substituted it into the model for calculation.

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MEMORANDUM

To: International Joint Commission(IJC)

From: Team # 2416805, ICM 2024

Subject: A strategy model for managing and controlling the Great Lakes

Date: February 6, 2024

Dear IJC leadership:

We are honored to describe our strategic model for managing the Great Lakes. The detailed information goes as follows.

First, we investigate the historical water level data of the Great Lakes in the past twenty years, as well as the flow of major rivers and the precipitation, evaporation, runoff and other related data of the Great Lakes. Based on these data and the real geographical location,we establish a network model of the Great Lakes, with lakes as points and rivers as edges. We relate the water levels of different lakes together through their water conservation equations and river flow formulas. In order to determine the optimal water levels of the lakes, we employ a multi-objective programming model to obtain the optimal water level of each lake in any month of the year. Thereafter, we developed PID feedback control algorithms for the compensating work(Soo Locks) and Moses-Sanders Dam to maintain optimal water levels.

Based on the in-depth analysis of the proposed models, we highlight the following strengths:

- Our model fully takes into account seasonal changes in environmental conditions such as precipitation, evaporation, and runoff, which is highly consistent with the actual situation.
- Our model considers the demands of various stakeholders and is accurate and practical
- Our model is on basis of a large amount of real data so that it is credible
- Our model is robust and stable and can mitigate abnormal changes in environmental conditions

All in all, we sincerely hope that our models and results could help you assist with management and models for the Great Lakes.

Yours Sincerely
Team # 2416805